CENG381 Stochastic Processes

Classifying States — Practice 1 (Answer Key)

Q1 Solution

a) Communicating classes:

State x communicates with y ($x \leftrightarrow y$) if each can reach the other.

{1, 2}: 1→2 (direct) and 2→1 (direct). Closed? From 1: can only reach 2 (and back). From 2: can only reach 1. No path to any other state. → Closed communicating class.

{3, 4}: 3→4 (prob 1/3) and 4→3 (prob 1). Closed? From 3: can escape to 5 or 6 (prob 2/3). → NOT closed (transient class).

{5}: absorbing singleton. Closed trivially.

{6}: absorbing singleton. Closed trivially.

b) Recurrent vs. transient:

States 1, 2: form a closed class → both recurrent.

States 5, 6: absorbing → recurrent (trivially).

States 3, 4: belong to a non-closed class. Once the chain escapes from {3,4} to {5} or {6} it never returns. P(ever return to 3 | start 3) < 1. → Both transient.

c) P(absorbed in 5 | start 3):

Let h = P(reach 5 | start at 3). Writing the balance equation:

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h = (1/3)·1  +  (1/3)·0  +  (1/3)·P(reach 5 | start 4)
  = 1/3  +  (1/3)·h
  [since from 4 the chain returns to 3 with probability 1]
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(2/3)·h = 1/3
h = 1/2
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By symmetry (equal probabilities 1/3 to each absorbing state or back to 4), the chain is equally likely to be absorbed in 5 as in 6. P(absorbed in 5 | start 3) = 1/2.

Q2 Solution

a) Irreducibility:

W→M (direct), M→W (direct), M→B (direct), B→M (direct). All states are reachable from each other.
→ Irreducible chain. Every state in an irreducible finite chain is recurrent.

b) Stationary distribution:

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Balance equations  π·P = π :
  π_W = (3/4)·π_W + (1/2)·π_M + 0·π_B       ... (I)
  π_M = (1/4)·π_W + 0·π_M   + 1·π_B          ... (II)
  π_B = 0·π_W   + (1/2)·π_M + 0·π_B          ... (III)
  π_W + π_M + π_B = 1                         ... (IV)
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From (I): (1/4)·π_W = (1/2)·π_M → π_M = (1/2)·π_W
From (III): π_B = (1/2)·π_M = (1/4)·π_W
From (IV): π_W + (1/2)π_W + (1/4)π_W = (7/4)π_W = 1

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π_W = 4/7,    π_M = 2/7,    π_B = 1/7
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c) Expected return time to W:

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E[T_W] = 1/π_W = 1/(4/7) = 7/4 = 1.75 days
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On average, after a working day, the production line returns to full operation in 1.75 days. (It will spend some days under maintenance or broken before returning to the Working state.)

Q3 Solution

a) Markov chain model:

States: {0, 1, 2, 3, 4} where k = number of distinct cards held.
State 4 is absorbing. Transition probabilities:

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From state k  (k < 4):
  P(k, k+1) = (4−k)/4   [pull a new card type]
  P(k, k)   = k/4       [pull a duplicate]

k=0: P(0,1)=1,   P(0,0)=0
k=1: P(1,2)=3/4, P(1,1)=1/4
k=2: P(2,3)=2/4, P(2,2)=2/4
k=3: P(3,4)=1/4, P(3,3)=3/4
k=4: P(4,4)=1
```

b) E[T] from 0 cards:

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E[time from k to k+1] = 4/(4−k):
  k=0 to 1:  4/4 = 1 step
  k=1 to 2:  4/3 steps
  k=2 to 3:  4/2 = 2 steps
  k=3 to 4:  4/1 = 4 steps

E[T] = 1 + 4/3 + 2 + 4 = 3 + 4/3 + 4 = 7 + 4/3 = 25/3
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Coupon−collector formula:  E[T] = 4·(1 + 1/2 + 1/3 + 1/4)
  = 4·(12/12 + 6/12 + 4/12 + 3/12)
  = 4·(25/12) = 100/12 = 25/3 ≈ 8.33  ✓
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c) P(all 4 cards after exactly 2 more purchases | currently have 2):

Starting at state 2. We need to be at state 4 after exactly 2 steps.

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Step 1 from state 2:
  P(go to 3) = 2/4 = 1/2
  P(stay at 2) = 2/4 = 1/2

P(reach 4 in exactly 2 steps | start 2):
  = P(2→3)·P(3→4) + P(2→2)·P(2→4 in 1 step)
  = (1/2)·(1/4) + (1/2)·0
  = 1/8
  [can't jump from 2 to 4 in one step]
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